

Investigation: Electron as a Vortex Ring with a Bound Aether Core

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April 2026 — Working Document

Abstract

We investigate a formation model for the electron in which a quantized vortex ring, created in the high-energy early universe, accumulates a bound Aether core through the Dynamic Casimir Effect (DCE). The bound ring grows until the core-boundary velocity drops below the DCE permanence threshold, at which point only transient binding (gravity) occurs and the electron mass stabilizes. Using the theory's fundamental parameters ($m_A \approx 500\text{--}850 \text{ MeV}/c^2$, $\rho_A \approx 4 \times 10^{11}\text{--}10^{12} \text{ kg/m}^3$), the self-consistent vortex equation predicts $m_e \approx 0.4\text{--}0.6 \text{ MeV}$, bracketing the observed 0.511 MeV . Including corrections for the core energy (+5–14%, model-dependent) and the Bernoulli density deficit near the vortex core (−1.7%), the prediction matches 0.511 MeV exactly for $m_A \approx 628\text{--}689 \text{ MeV}/c^2$, well within the independently constrained convergence zone.

1 Formation Narrative

The electron forms through the following sequence in the early universe:

1. **Vortex ring creation.** In the high-energy early universe, turbulent Aether flows create quantized vortex rings with circulation $\kappa_e = h/(2m_A)$ —the minimum (spin- $\frac{1}{2}$) quantum. Initially these are pure fluid vortices: no bound material, no core structure beyond the healing-length depletion.
2. **DCE creates matter.** The circulation velocity at the healing length is $v(\xi) = c/\sqrt{2} \approx 0.707 c$ (Eq. 2). This exceeds the DCE permanence threshold, so the rapidly-circulating boundary continuously converts vacuum fluctuations into permanent bound Aether.
3. **Matter migrates inward.** The Bernoulli pressure in a vortex is lowest at the core (Eq. 3). Bound material, no longer part of the superfluid flow, is pushed inward by the pressure gradient and accumulates along the vortex tube axis—forming a toroidal ring of bound Aether tracing the vortex's major radius r_{loop} .
4. **Ring grows.** As more bound material falls to the center, the ring expands radially outward from the tube axis. The effective core boundary moves from ξ to a larger radius r_{eq} .

5. **Boundary velocity drops.** The circulation κ_e is quantized and fixed, but the velocity at the boundary is $v(r) = \kappa_e/(2\pi r)$. As r_{eq} increases, the boundary velocity decreases.
6. **Permanence threshold reached.** When $v(r_{\text{eq}})$ drops to v_{perm} —the minimum velocity for permanent DCE binding—the vortex can no longer create permanent mass. Only transient bind–unbind cycling occurs at the boundary, producing the pressure asymmetry that manifests as gravity (Paper 3).
7. **Mass stabilizes.** The electron’s mass is the sum of the circulation kinetic energy and the bound ring’s rest mass at this equilibrium point.

This is a *self-regulating* process: the creation of mass enlarges the core, which reduces the boundary velocity, which eventually shuts off mass creation. The electron mass is not a free parameter but an equilibrium outcome determined by medium properties.

Energy source. Where does the ring’s rest-mass energy come from? The DCE mechanism converts energy from the vortex’s circulation into bound material—the moving boundary does work on the vacuum fluctuations, and the energy is drawn from the kinetic energy of the flow. As the ring grows, the circulation KE decreases by the same amount that the ring mass increases. The total energy (KE + ring mass) is therefore approximately *conserved* during the formation process: $m_e \approx E_{\text{KE,initial}} = E_{\text{KE,final}} + m_{\text{ring}} c^2$. The electron’s final mass is set by the initial vortex energy at creation, not by the partitioning between KE and ring mass.

2 Key Equations

2.1 Circulation Velocity

For a quantized vortex with circulation $\kappa_e = h/(2m_A) = \pi\hbar/m_A$, the flow velocity at distance r from the tube axis is:

$$v(r) = \frac{\kappa_e}{2\pi r} \quad (1)$$

At the healing length $\xi = \hbar/(m_A c \sqrt{2})$:

$$v(\xi) = \frac{\pi\hbar/m_A}{2\pi \hbar/(m_A c \sqrt{2})} = \frac{c}{\sqrt{2}} \approx 0.707 c \quad (2)$$

2.2 Bernoulli Pressure

The pressure distribution in the vortex flow:

$$P(r) = P_\infty - \frac{1}{2} \rho_A v^2(r) = P_\infty - \frac{\rho_A \kappa_e^2}{8\pi^2 r^2} \quad (3)$$

Pressure is lowest at the core center ($r \rightarrow 0$) and increases outward. Bound material (not participating in superfluid flow) experiences a net inward force from the surrounding higher-pressure fluid.

2.3 Equilibrium Condition

The bound ring grows until the boundary velocity reaches the permanence threshold:

$$v(r_{\text{eq}}) = v_{\text{perm}} \implies r_{\text{eq}} = \frac{\kappa_e}{2\pi v_{\text{perm}}} \quad (4)$$

For $v_{\text{perm}} < c/\sqrt{2}$, the ring extends beyond the healing length: $r_{\text{eq}} > \xi$.

2.4 Core Energy: Already Accounted For

An important subtlety: does the Donnelly formula already include the core’s energy?

The Donnelly formula computes only the *kinetic energy of the circulating flow outside the core*. In a standard GPE vortex, the total energy also includes the core’s gradient and potential energy—the cost of depleting the condensate. This core energy is of order $1/\ln(r_{\text{loop}}/\xi)$ of the kinetic energy. For the electron’s aspect ratio, this is $\sim 5\text{--}14\%$ depending on the condensate potential (the exact value for the logarithmic potential used in this theory has not been computed from first principles).

In the bound-core model, the core region is filled with bound material instead of depleted vacuum. The bound ring’s rest mass *replaces* the gradient and potential energy of the depleted core—it is the physical realization of the same core energy, not an additional contribution on top of it. The total electron energy is therefore:

$$m_e c^2 \approx E_{\text{KE}}(\text{circulation}) + E_{\text{core}} \quad (5)$$

where E_{core} exists in both models (depleted or bound) and is $\sim 5\text{--}14\%$ of E_{KE} . In the depleted model it is gradient + potential energy; in the bound model it is the ring’s rest mass. The Donnelly formula (which captures only E_{KE}) undercounts the total by this amount regardless of core interpretation.

2.5 Bernoulli Density Deficit Near the Core

The Donnelly formula assumes constant density ρ_A everywhere. But the vortex circulation itself produces a Bernoulli density deficit (fixes.txt, Fix 4.1: “venturi-like deficit from driven spin energy”):

$$\frac{\delta\rho(r)}{\rho_A} \approx -\frac{v^2(r)}{2c^2} = -\frac{\kappa_e^2}{8\pi^2 c^2 r^2} \quad (6)$$

At the healing length ($r = \xi$, $v = c/\sqrt{2}$), this gives $\delta\rho/\rho_A \approx -25\%$ —not negligible. The density drops rapidly with distance:

r/ξ	v/c	$\delta\rho/\rho_A$
1.0	0.707	−25%
1.5	0.471	−11%
2.0	0.354	−6%
5.0	0.141	−1%

The density deficit reduces the kinetic energy because $E_{\text{KE}} \propto \rho(r) v^2(r)$. The correction is $\delta E = \int \frac{1}{2} (-\delta\rho) v^2 dV$, giving:

$$\frac{\delta E_{\text{KE}}}{E_{\text{KE}}} \approx -\frac{1}{8 \ln(r_{\text{loop}}/\xi)} \approx -1.7\% \quad (7)$$

(The factor 1/8 arises because $\kappa_e^2/(16\pi^2 c^2 \xi^2) = 1/8$ for $v(\xi) = c/\sqrt{2}$.) Note: $\delta\rho/\rho_A$ is a genuine density change (not just a pressure change with constant ρ_A) because the vortex circulation is *driven* flow, not free-fall—the persistent spin energy maintains the flow against Bernoulli pressure, creating a venturi-like deficit. This is physically distinct from gravitational inflow, where free-fall dynamics ensure $\Delta\rho = 0$ exactly (Paper 2). For the logarithmic equation of state, the exact compressible Bernoulli gives $\rho(r) = \rho_A e^{-v^2/(2c^2)}$ rather than the linearized $\rho_A [1 - v^2/(2c^2)]$; the integrated KE correction differs by less than 3% between exact and linear forms, so the linearized result -1.7% is adequate.

The Donnelly formula itself is non-relativistic. In the Aether framework, relativistic effects emerge from the medium at $O(v^2/c^2)$: time dilation from wave path geometry, length contraction from density changes (Paper 2). The Bernoulli density correction is the leading such correction to the vortex energy; higher-order terms are $O(v^4/c^4) \sim 0.4\%$, negligible at this level of precision.

2.6 Electron Mass

Combining all contributions:

$$m_e c^2 \approx \underbrace{\frac{1}{2} \rho_A \kappa_e^2 r_{\text{loop}} \left[\ln\left(\frac{8 r_{\text{loop}}}{r_{\text{eq}}}\right) - \frac{1}{2} \right]}_{E_{\text{KE}} \text{ (circulation)}} \times \underbrace{(1 - 0.017)}_{\text{Bernoulli}} + \underbrace{E_{\text{core}}}_{\sim 5-14\% \text{ of } E_{\text{KE}}} \quad (8)$$

The first term is the Donnelly formula corrected for the Bernoulli density deficit (-1.7%). The second is the core energy (5–14% of E_{KE}), which in the bound-core model is the ring's rest mass. The net correction from the bare Donnelly formula is:

$$m_e c^2 \approx E_{\text{KE}}^{\text{Donnelly}} \times (1 - 0.017 + f_{\text{core}}) \quad (9)$$

where $f_{\text{core}} \approx 0.05\text{--}0.14$ is the core energy fraction (model-dependent). The net correction ranges from $+3\%$ to $+12\%$ above the bare Donnelly value.

The loop radius is set by the quantum minimum: $r_{\text{loop}} = \hbar/(m_e c)$ (the reduced Compton wavelength).

3 Mass Estimate

3.1 Circulation KE as a Lower Bound

As a first step, ignore the ring mass and set $m_e c^2 = E_{\text{KE}}$ alone. Using the self-consistent condition $r_{\text{loop}} = \hbar/(m_e c)$ and setting $r_{\text{eq}} = \xi$:

$$m_e^2 c^3 = \frac{1}{2} \rho_A \kappa_e^2 \hbar \left[\ln\left(\frac{8 r_{\text{loop}}}{r_{\text{eq}}}\right) - \frac{1}{2} \right] \quad (10)$$

$$m_e^2 = \frac{\pi^2 \rho_A \hbar^3}{2 m_A^2 c^3} \left[\ln \left(\frac{8\sqrt{2} m_A}{m_e} \right) - \frac{1}{2} \right] \quad (11)$$

This gives a *lower bound* on m_e (since it ignores the ring's mass). Solving iteratively for $\rho_A = 5 \times 10^{11} \text{ kg/m}^3$:

m_A (MeV/ c^2)	ξ (fm)	r_{loop} (fm)	m_e (KE only) (MeV)	vs. observed
500	0.28	325	0.61	+19%
550	0.25	354	0.56	+9%
600	0.23	383	0.515	+1%
650	0.21	411	0.48	−6%
700	0.20	439	0.45	−12%
800	0.17	495	0.40	−22%

The bare KE-only equation gives $m_e = 0.515 \text{ MeV}$ at $m_A = 600$. Applying the net correction from Sections 2.4–2.5 (+3% to +12%, depending on core energy) increases the prediction. Matching 0.511 MeV requires m_A above 600 MeV (see below).

3.2 Full Prediction with Energy Conservation

Since DCE converts circulation KE into ring mass (energy conserved), the total electron mass is set by the initial vortex energy, not by the ring's size. The self-consistent equation remains Eq. (11)—but now m_e represents the *total* energy (KE + ring), and the Donnelly formula gives only the KE portion.

Including the core energy correction and Bernoulli deficit from Sections 2.4–2.5, the corrected prediction depends on the core energy fraction f_{core} :

f_{core}	Net correction	Best-fit m_A	KE only (MeV)	Corrected (MeV)
5%	+3%	628 MeV	0.495	0.511
8%	+6%	649 MeV	0.480	0.510
11%	+9%	669 MeV	0.468	0.512
14%	+12%	689 MeV	0.455	0.511

For any core energy in the 5–14% range, there exists an m_A in the convergence zone (500–850 MeV/ c^2) that matches the observed mass:

$m_e \approx 0.511 \text{ MeV}$ (predicted, $m_A \approx 628\text{--}689 \text{ MeV}$)	vs.	$m_e = 0.511 \text{ MeV}$ (observed)
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(12)

The match is exact for m_A values well within the convergence zone. The Bernoulli density deficit (−1.7%) partially offsets the core energy (+5–14%), and the remaining net correction (+3–12%) is absorbed by m_A , which shifts from 600 MeV (bare Donnelly) to 628–689 MeV (corrected).

3.3 Effect of Threshold Velocity

Under energy conservation, the total electron mass is fixed by the initial vortex energy at creation. The threshold velocity v_{perm} determines only the *partitioning* between circulation KE and ring rest mass, not the total. As v_{perm} decreases (ring grows), more KE converts to ring mass, but m_e stays constant.

The ring mass at equilibrium is:

$$m_{\text{ring}} c^2 = f_{\text{core}} \times m_e c^2 \quad (13)$$

where $f_{\text{core}} \approx 5\text{--}14\%$ (the fraction of total energy in the core, see Section 2.4). This is a small fraction: the electron remains circulation-dominated at equilibrium.

4 Physical Structure at Equilibrium

4.1 Compression Ratio

Even if the entire electron mass were packed into the core ring (the extreme $f = 1$ case), the compression ratio would be remarkably low:

$$X = \frac{\rho_{\text{core}}}{\rho_A} = \frac{m_e}{\rho_A \cdot 2\pi^2 r_{\text{loop}} \xi^2} \approx 4\text{--}5 \quad (14)$$

(varying from ~ 4.4 at $m_A = 600$ to ~ 4.9 at $m_A = 630$). Only ~ 5 times ambient density even in the extreme case—far below the $\sim 7 \times 10^7$ compression in QCD bridges. This reflects the electron’s small mass (0.511 MeV) spread over a large circumference ($2\pi r_{\text{loop}} \approx 2400$ fm).

For the core energy range $f_{\text{core}} = 5\text{--}14\%$, the ring carries only 0.03–0.07 MeV. Spread over the core volume ($\sim 300\text{--}400$ fm³), the ring density is well below ρ_A —the core is sparsely filled, not compressed.

4.2 What Stops the Growth

The self-regulation mechanism works as follows:

1. DCE creates bound material at the core boundary where $v > v_{\text{perm}}$
2. Material falls inward, expanding the effective core from ξ to r_{eq}
3. At r_{eq} , the velocity has dropped: $v(r_{\text{eq}}) < v(\xi)$
4. When $v(r_{\text{eq}}) = v_{\text{perm}}$, permanent binding ceases
5. Transient binding continues (producing gravity), but no new mass accumulates

The quantized circulation $\kappa_e = h/(2m_A)$ is topologically fixed—it cannot change. Only the boundary radius changes as the ring grows, reducing the velocity through the $v \propto 1/r$ dependence.

5 The DCE Permanence Threshold

The permanence threshold v_{perm} is the single most important unknown in this model. It determines:

- The equilibrium core size $r_{\text{eq}} = \kappa_e / (2\pi v_{\text{perm}})$
- The partitioning between circulation KE and bound ring mass
- The detailed structure of the electron

For v_{perm} near $c/\sqrt{2}$ (modest ring growth), the total mass varies only weakly with v_{perm} : the logarithmic KE decrease partially cancels the ring mass increase. For $v_{\text{perm}} \ll c$, however, the ring mass (scaling as r_{eq}^2) dominates the logarithmic KE decrease, and the total mass grows without bound. This constrains v_{perm} to be close to $c/\sqrt{2}$.

The threshold must satisfy $v_{\text{perm}} < c/\sqrt{2}$ (otherwise no ring growth occurs) and $v_{\text{perm}} > 0$ (otherwise the ring grows without limit). A natural estimate from the Schwinger pair-production analogy—where the creation rate transitions from exponentially suppressed to significant when the exponent $(c/v)^2 \sim \pi$ —gives:

$$v_{\text{perm}} \sim \frac{c}{\sqrt{\pi}} \approx 0.56 c \quad (15)$$

This corresponds to $r_{\text{eq}} \approx 1.26 \xi$ —a modest expansion of the core beyond the healing length.

Quantifying v_{perm} from first principles (the DCE pair creation rate in vortex geometry) is the key open problem identified in Paper 7.

6 Significance

6.1 Electron Mass from Medium Properties

The central result is Eq. (11) combined with Eq. (13): the electron mass is determined by m_A , ρ_A , c , and the ring density α —all properties of the Aether medium and the equilibrium vortex structure, not free parameters of the electron. The mass arises from:

- **Circulation quantization:** $\kappa_e = h/(2m_A)$ fixes the flow speed at each radius
- **Quantum minimum size:** $r_{\text{loop}} = \hbar/(m_e c)$ prevents the vortex ring from shrinking to zero
- **Self-consistency:** m_e must equal the energy of a vortex ring of size r_{loop} , which itself depends on m_e

For $m_A \approx 628\text{--}689$ MeV (within the convergence zone), the corrected prediction matches the observed 0.511 MeV exactly. The inputs— m_A and ρ_A —are independently constrained by QCD bridge physics (Paper 5) and the healing-length convergence (Paper 7). The core energy fraction f_{core} is the remaining model-dependent quantity; computing the vortex core structure in the logarithmic potential would pin down both f_{core} and the best-fit m_A simultaneously.

6.2 Self-Regulating Mass

The bound-core model explains *why* the electron has the mass it does: the DCE mechanism creates mass until it shuts itself off. Every electron reaches the same equilibrium because every minimum-circulation vortex ring has the same κ_e , the same velocity profile, and therefore the same threshold crossing point. Mass universality follows from circulation quantization.

6.3 Open Tube / Closed Tube Taxonomy

The bound electron core ring is a *closed tube* of gently compressed Aether, stabilized by the encircling vortex circulation. The QCD bridge is an *open tube* of strongly compressed Aether, stabilized by quark vortex endpoints. Both are topologically-stabilized bound Aether structures in the same medium, differing in topology (closed vs. open), compression ($\ll 1$ to ~ 5 vs. $\sim 7 \times 10^7$), and energy scale (0.5 MeV vs. ~ 1 GeV/fm).

7 Open Questions

1. **DCE permanence threshold.** What is v_{perm} ? A first-principles DCE calculation in vortex geometry would determine the equilibrium core size and the KE/ring mass partitioning.
2. **Muon and tau.** Could higher-energy vortex configurations (multiply-wound cores, knotted rings) stabilize at different masses? The self-regulation mechanism would apply to any circulation quantum, potentially producing the lepton mass hierarchy.
3. **Form factor.** Electron scattering experiments probe to $\sim 10^{-18}$ m with no structure seen, yet the vortex model predicts structure at $r_{\text{loop}} \sim 400$ fm. How does the vortex ring's scattering cross-section mimic a point particle?
4. **Charge origin.** The bound core ring surface is a permanent DCE boundary. Could the intense transient cycling at this surface be the microscopic mechanism that implements the radial Aether flow defining electric charge (an acknowledged gap in Paper 4)?
5. **Cosmological formation.** What conditions in the early universe create vortex rings with the minimum circulation quantum? What determines the initial r_{loop} , and how does it relax to $\hbar/(m_e c)$?